

Mock Lab Test — Round 2

This one is designed with theory-heavy, tricky statements, short code, calculations, and lots of **checkbox questions**.

Marking

- **Q1–Q10:** One correct answer → **1 mark each**
- **Q11–Q20:** Select **ALL correct answers**
 - **+1** for each correct selection
 - **–1** for each incorrect selection
 - Minimum score per question = **0**

Don't look at your notes. Answer like the real exam. 😊

Part A — Single Answer

Q1. Train/Test Split

You have a dataset with **500 samples**:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2
)
```

How many samples are expected in the **training set**?

- A. 100 B. 200 C. 300 D. 400

Q2. StandardScaler

Which situation is the **best use case** for StandardScaler?

- A. Converting "Male" and "Female" into numbers
B. Making numerical features have approximately mean 0 and standard deviation 1
C. Removing missing values
D. Balancing minority classes

Q3. Nominal Data

Which encoding method is generally most suitable for a **nominal categorical variable**?

Example:

Country = Sri Lanka, India, Japan

- A. Label Encoding
B. One-Hot Encoding
C. StandardScaler
D. PCA

Q4. Correlation

A scatter plot shows that as X increases, Y also generally increases, but the points have noticeable variation and do not lie exactly on a straight line.

This indicates:

- A. Perfect positive correlation
B. Positive correlation, but not perfect
C. Perfect negative correlation
D. No correlation

Q5. Outliers

Which plot is usually most suitable for identifying outliers?

- A. Box plot
B. Pie chart
C. Bar chart
D. Histogram only

Q6. Euclidean Distance

Calculate the Euclidean distance between:

$P = (2, 3)$ and **$Q = (5, 7)$**

- A. 3
B. 4
C. 5
D. 7

Q7. Logistic Regression

What does the following equation represent?

$$z = Xw + b$$

- A. The sigmoid function
 - B. The linear combination before applying sigmoid
 - C. The Log Loss function
 - D. Euclidean distance
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Q8. PCA

PCA is mainly used to:

- A. Increase the number of samples
 - B. Reduce dimensionality while preserving important variance
 - C. Remove all outliers
 - D. Balance the dataset
-

Q9. Model Performance

A model achieves:

Training accuracy = 98%

Testing accuracy = 60%

This most likely indicates:

- A. Underfitting
 - B. Good fit
 - C. Overfitting
 - D. Data normalization
-

Q10. Gradient Descent

Consider:

$$\theta := \theta - \alpha \nabla J(\theta)$$

What does α represent?

- A. Number of features
 - B. Learning rate
 - C. Test size
 - D. Number of neighbors
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Part B — Multiple Answer / Checkbox Questions

Q11. Data Balancing

Select **ALL** techniques that can address class imbalance.

- ☐ A. Random oversampling
 - ☐ B. Random undersampling
 - ☐ C. SMOTE
 - ☐ D. One-Hot Encoding
 - ☐ E. Increasing the minority class representation
-

Q12. SMOTE vs Scaling

Select all correct statements.

- ☐ A. SMOTE is used to help address class imbalance.
 - ☐ B. SMOTE creates synthetic examples of the minority class.
 - ☐ C. StandardScaler is primarily used to create new minority-class samples.
 - ☐ D. Scaling changes the numerical scale of features.
 - ☐ E. SMOTE and MinMaxScaler perform the same task.
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Q13. Data Leakage

Two people preprocess data differently.

Person P

1. Standardize the complete dataset
2. Split into training and testing sets

Person Q

1. Split into training and testing sets
2. Fit StandardScaler on X_train
3. Transform X_train and X_test

Select all correct statements.

- ☐ A. Person Q follows the correct general preprocessing approach.
- ☐ B. Person P can cause data leakage.
- ☐ C. Person P's test data influences the scaling parameters.

- ☐ D. Person Q should fit a separate scaler independently on X_{test} .
 - ☐ E. Person Q should use the scaler fitted on training data to transform X_{test} .
-

Q14. Stratified Splitting

Select all correct statements.

- ☐ A. Stratified splitting attempts to preserve class proportions.
 - ☐ B. It can be especially useful for imbalanced classification datasets.
 - ☐ C. It guarantees that every numerical feature has the same mean in train and test sets.
 - ☐ D. In Scikit-learn, `stratify=y` can be used with `train_test_split()`.
 - ☐ E. It randomly removes duplicate rows.
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Q15. PCA and Feature Scale

You have:

Feature 1 $\rightarrow$ 0 to 10

Feature 2 $\rightarrow$ 0 to 100,000

Select all correct statements.

- ☐ A. Feature 2 may dominate because of its much larger scale.
 - ☐ B. Standardizing features before PCA is often appropriate.
 - ☐ C. PCA automatically ignores Feature 2.
 - ☐ D. PCA is related to identifying directions/components that capture variance.
 - ☐ E. PCA always preserves 100% of the original variance after reducing dimensions.
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Q16. Gradient Descent Calculation

Suppose:

$$\theta = 10$$

$$\alpha = 0.1$$

$$\frac{\partial J}{\partial \theta} = 4$$

Using:

$$\theta_{\text{new}} = \theta - \alpha \frac{\partial J}{\partial \theta}$$

Which statements are correct?

- ☐ A. The update is $10 - (0.1 \times 4)$.
 - ☐ B. The new value of θ is 9.6.
 - ☐ C. The new value of θ is 10.4.
 - ☐ D. Gradient descent moves in the opposite direction of the gradient.
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Q17. Underfitting, Good Fit and Overfitting

Select all correct statements.

- ☐ A. Underfitting may produce poor performance on both training and testing data.
 - ☐ B. Overfitting can produce very high training performance but poor testing performance.
 - ☐ C. A good-fit model must have 100% accuracy on the training set.
 - ☐ D. A model that is too simple may underfit.
 - ☐ E. Overfitting always means the model has low training accuracy.
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Q18. kNN

Select all correct statements.

- ☐ A. kNN uses nearby data points to make predictions.
 - ☐ B. Euclidean distance can be used to measure closeness.
 - ☐ C. A small value of K can be sensitive to noise.
 - ☐ D. A very large K may lead to oversimplification/underfitting.
 - ☐ E. kNN always uses sigmoid activation.
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Q19. Odd K

Select all correct statements.

- ☐ A. An odd K can help reduce voting ties in binary classification.
 - ☐ B. K must always be an odd number.
 - ☐ C. With binary classes and $K = 4$, a 2–2 voting tie is possible.
 - ☐ D. Odd K guarantees the best possible model accuracy.
 - ☐ E. K represents the number of nearest neighbors considered.
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Q20. Correlation Analysis

Select all correct statements.

- ☐ A. A positive correlation means two variables generally tend to increase together.
- ☐ B. A negative correlation means one variable tends to increase while the other tends to decrease.
- ☐ C. Correlation always proves causation.
- ☐ D. A correlation close to +1 indicates a strong positive linear relationship.
- ☐ E. A correlation close to 0 necessarily proves there is no relationship of any kind.